

Paridhi Latawa

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EDUCATION

- Massachusetts Institute of Technology (MIT)**, M.Eng. in EECS (CS + Molecular Biology) Cambridge, MA
• Master's Thesis in progress with Feng Zhang Lab. May 2026
- Massachusetts Institute of Technology (MIT)**, B.S. in EECS (CS + Molecular Biology) Cambridge, MA
• Minor in Economics. NEET Living Machines Certificate. May 2026
- Liberal Arts and Science Academy** Austin, TX
• Regeneron STS Scholar, Coca-Cola Scholar, Apple Scholar May 2022

COURSEWORK

- **Graduate:** ML for Comp Bio, Gen AI for Biology, AI/Decision Making for Medicine, AI for Protein Biology
- **Undergraduate CS/Math:** Natural Language Processing, Machine Learning, Probability & Statistics, Algorithms, Linear Algebra, Programming, Discrete Math, Calculus I & II
- **Undergraduate Biology:** Biochemistry, Cell Biology, Organic Chemistry, Thermodynamics, Genetics

TEACHING AND MENTORSHIP EXPERIENCE

- Massachusetts Institute of Technology** – Teaching Assistant (*Intro to Probability & Statistics*) Feb 2025 – May 2025
• Led weekly in-class problem-solving sessions, office hours, & coding lessons (140+ students); Graded HW.
- Massachusetts Institute of Technology** – Associate Advisor Aug 2025 – Present
• Advised students with academic planning, course selection, & research involvement.
• Collaborated with Professor Collin Stultz to improve advisee support and registration processes.

RESEARCH EXPERIENCE

- Broad Institute of MIT & Harvard, Feng Zhang Lab** - Researcher Sep 2022 – Present
• Project 1: Investigated bacterial antiviral defense system proteins using bacterial growth, PCR, & cloning.
• Project 2: Conducted computational & wet-lab mutagenesis of DRT reverse transcriptase bacterial system. Used software/bioinformatics tools like Geneious & Benchling to ID essential sequence regions for system's function.
• Project 3: Led an independent computational study on characterizing the evolutionary diversity of the L7-protein domain using FoldSeek, DALI, AlphaFold, iTOL, & embedding visualization
- Koch Institute for Integrative Cancer Research, Langer/Traverso Lab** - Researcher Sep 2023 – Present
• Project 1: Helped build a framework to integrate biology specific foundation model (ESM, UniMol) knowledge into general LLMs using latent representations. Set up HPC/datasets & iterated through initial implementations.
• Project 2: Helped develop a *de novo* molecular design algorithm based on physiologically relevant constraints. Tested effects of jointly constraining generation with multiple properties.
• Project 3: Supported development of QA-Token, an RL-based quality-aware tokenization approach. Contributed to the evaluation of the approach in the NLP text domain.
- Eli Lilly and Company** - AI and Technology Intern May 2025 – Aug 2025
• Deployed end-to-end ML fine-tuning pipeline for in silico single-cell perturbation prediction.
• Presented project in poster symposium, technical talk, & presentation to 70+ company stakeholders & senior VPs.
- MD Anderson Cancer Center, Bin Wang Lab** - Research Intern May 2024 – Aug 2024
• Researched DNA repair for Abraxas-BRCA-1 complex via western blotting, IF, FISH, & microscopy.
• Spearheaded collaboration with the Genetics Department Microscopy Core Director & the Imaris Imaging team.
- Dianox (AI/Biotech startup); MIT MISTI Program** – Research Intern May 2023 – Aug 2023
• Identified aptamer-based drug targets using ML & fluorescence wet lab techniques.
• Led analysis of Danish innovation ecosystem with 11+ experts at Novo Nordisk, US Embassy, & Uni of Cph.

PUBLICATIONS

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|---|------|
| [1] Can LLMs Reason Over Non-Text Modalities in a Training-Free Manner? A Case Study with In-Context Representation Learning | 2025 |
| Tianle Zhang, Wanlong Fang, Jonathan Woo, Paridhi Latawa , Deepak Subramanian, Alvin Chan
<i>The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS)</i> | |
| [2] Guiding Diffusion-Based Generation with Experimental and Physicochemical Data: Constraint Latent Space Mapping and Task Arithmetic for Multi-Constrained Molecular Generation | 2025 |
| Deepak Subramanian, Rowan Honeywell, Sophia Yao, Paridhi Latawa , Giovanni Traverso.
<i>2025 Molecular Machine Learning Conference [Poster Presenting Author]</i> . | |
| [3] Learning What Makes a Good Token: Quality-Aware Tokenization via Adaptive Reinforcement Learning for Noisy Sequential Data | 2025 |
| Arvid Gollwitzer, Paridhi Latawa , David de Gruijl, Deepak Subramanian, Giovanni Traverso
<i>The Fourteenth International Conference on Learning Representations (ICLR)</i> , Under Review | |
| [4] AptABERT: Predicting aptamer binding interactions | 2023 |
| Flemming Morsch, Iswarya Umasankar, Lys Moreta, Paridhi Latawa , Danny Lange, Jesper Wengel, Hiram Konjen, Christian Code, <i>Preprint</i> , doi.org/10.1101/2023.11.24.568626 | |
| [5] A Computational Modeling Framework to Analyze Synovial-Tissue Based Drug Targets and Diagnostic Biomarkers in Rheumatoid Arthritis | 2021 |
| Paridhi Latawa , Briana Chrisman, <i>Preprint</i> , doi.org/10.1101/2021.11.08.467813 | |
| [6] Applications of Convolutional Codes to DNA Codes and Error-Correction | 2021 |
| Paridhi Latawa , Dr. Nuh Aydin, <i>Preprint</i> , doi: 2111.04133v1 . | |
| [7] Identification & Analysis of Novel Synovial Tissue-Based Biomarkers & Interacting Pathways for Rheumatoid Arthritis | 2021 |
| Paridhi Latawa , <i>Preprint</i> , doi.org/10.1101/2021.03.19.21253995 | |
| [8] A Comparative Analysis of COVID-19 Vaccine Distribution Efforts in India and Mongolia through Data Visualization - Paridhi Latawa, et al., | 2021 |
| <i>Submitted to Journal of Emergency Medicine</i> | |

RESEARCH SYMPOSIUMS/TALKS

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| [1] In Silico Perturbation Prediction Modeling using Single Cell Foundation Models | 2025 |
| Paridhi Latawa , <i>Eli Lilly Advanced Intelligence - Engineering & Biology Research, Presentation, Talk</i> | |
| [2] Exploring the Functional Role of the L7-like Ribosomal Protein | |
| Paridhi Latawa , Guilhem Faure, Feng Zhang - <i>MIT EECS SuperUROP Poster Symposium</i> | |
| [3] Elucidating the Role of Abraxas on Telomere Maintenance & Alternative Lengthening of Telomeres | 2024 |
| Paridhi Latawa , Longqiang Wang, Bin Wang
<i>MD Anderson Cancer Center CATALYST Poster Symposium & Division of Radiation Oncology Symposium</i> | |
| [4] Investigating the Capabilities of Large Language Models for Protein Engineering | 2024 |
| Paridhi Latawa , Alvin Chan, Giovanni Traverso, Robert Langer
<i>MIT Biology Undergraduate Research Symposium</i> | |

MIT ACADEMIC COURSE PROJECTS

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| [1] Conditional Generation from Few-shot Examples for Protein Design | 2025 |
| Zhilei Bei, Weilang Luo, Paridhi Latawa . <i>6.s981: AI for Protein Biology (In Progress)</i> | |
| [2] Combining Sentiment, Topic Modeling, & a ChatBot for Social Data Analysis | 2024 |
| Paridhi Latawa , Anika Puri, Sejal Rathi. <i>6.8611: Quantitative Methods for Natural Language Processing</i> | |
| [3] A Pseudotime Analysis of Cellular Trajectories in Alzheimer's Disease using scRNA Sequencing | 2024 |
| Paridhi Latawa , Jenny Prince, Selina Qian. <i>6.8701: ML for Computational Biology</i> | |
| [4] Encoder-Decoder Model for Multimodal SMILES Latent Space Representation Property Prediction. | 2024 |
| Vincent Lin, Paridhi Latawa , Vanessa Xiao. <i>6.8711: Generative AI for Biology</i> | |
| [5] An Investigation of the Biotechnology and Life Sciences Innovation Ecosystems in Denmark | 2023 |
| Paridhi Latawa . <i>MIT Innovation Diplomats Program</i> | |

HONORS AND AWARDS

MIT Eric & Wendy Schmidt Center Research & Innovation Scholar (Feng Zhang Lab) Aug 2024 – May 2025

- [MIT EECS SuperUROP Scholar](#): Selected as 1 of 8 computational biology scholars as part of larger cohort.

HackMIT WisprAI Track 3rd Place

2025

NCWIT Award for Aspirations in Computing (3x National Awardee-top 9.5% in US), Impact Award 2018-Present

- MIT Campus Representative. Led 4+ colleges to organize multiple NCWIT AiC Boston-area receptions with 100+ attendees from 9 colleges & 11 High Schools - 2024 [Event Report](#), 2025 [Event Report](#)
- Recognized with the Impact Award for leadership in expanding access to computing

AnitaB Org Grace Hopper Conference & SWE WE24 Conference Grant Awardee & Ambassador

2024

Regeneron International Science & Engineering Fair Finalist, Texas Science & Engineering Fair

2022

Austin Regional Science Fair: 2022: Best of Fair Award. 2021: Superintendents Award

2021-2022

World Science Scholar | Bank of America Student Leader | Texas Education Agency Student Hero

LEADERSHIP AND COMMUNITY SERVICE

Co-Founder, EasyComm - startup funded & supported by MIT Sandbox Innovation Fund

Aug 2022 - Present

- Developing innovative speech technology app for children with Down Syndrome. **Patent pending.**
- Selected for Harvard Law Entrepreneurship Project/Cooley LLP program & Takeda Biopharma Internship
- Presented @ MIT Founder's Journey Advising Seminar, Nuts & Bolts of New Ventures Seminar, Health Equity Venture Capital Talk & Sandbox Showcases talks.

MIT School of Engineering Student Advisory Group, EECS Comm. Engagement Committee

Sep 2024 – Present

- Connect with Dean & faculty to experiences & perspectives on education, engagement, research, & student life. Created proposals for alumni funding requests for national conferences.

Lead Communications Ambassador, MIT Priscilla King Gray (PKG) Center for Public Service Sep 2023 - Present

- Facilitate connection between PKG & students & led 2 interns. Organized outreach & media presence.

Board Member, MIT South Asian Association of Students

Sep 2023 – Present

- Organized the largest cultural events at MIT, such as Garba, Holi, & South Asian Formal

MIT Innovation Diplomats (iDiplomats) Program

May 2023 – Sep 2023

- Researched & analyzed Denmark's biotech innovation ecosystem. Interviewed 11 leaders from government, industry, & academia; Authored a report on innovation & entrepreneurship.

Developmental and Behavioral Pediatrics, Boston Medical Center

Dec 2022 – Feb 2023

- Supported 8+ low-socioeconomic families & patient cases with Autism Spectrum Disorder

Founder and Executive Director, SparkED – STEM Nonprofit Organization.

Apr 2020 – Dec 2022

- Equipped 2000+ youth across 11 countries globally with STEM resources. Led 20+ volunteer team.

Social Chair & Member, MIT Laureates and Leaders Program, Office of Minority Education

Sep 2024 – Present

Recruitment Director, Panhellenic Delegate, MIT Panhellenic Council

May 2023 – Present

MIT Undergraduate Association Education Committee, Community Service Committee

Aug 2022 – May 2023

SKILLS AND INTEREST

Interests: Rowing, hiking, biking, videography, baking, painting, viola, puzzles, science communication, blogging

Language Skills: Advanced in English, Hindi, Punjabi; Beginner in Spanish

Computational Skills: ML, LLMs, Python, Torch, Pandas, Biopython, R Studio, Linux, HPC, ImageJ/Fiji, Imaging

Wet Lab Skills: PCR, Gel Electrophoresis, Western Blot, IF, FISH, Protein Purification, Transformation, Gibson

Assembly, Fluorescence Anisotropy/Polarization, Fluorescence Microscopy; Biotech Lab Technician Certificate (ACC)